Operators: Operator is a special symbol is used do a particular operation [work]. Operator is working on operands. C comes with 44 operators and 14 separators.

Based on no of operands participating in operation, the operators divided into 3 types.

1. **Unary operators**: Require one operand.
   Eg: **a**++, **a**--, ++**a**, --**a**, sizeof(**a**), +**a**, -**a**, ~**a**, !**a**,..
2. **Binary operators**: Require two operands.
   Eg: a+b, a-b, a>b, a<=b, a==b, a!=b, a<<b, a>>b,.
3. **Ternary operator / conditional op [? :]**: It require 3 operands.
   Eg: **conditional part ? true part : false part ;**

Based on operation, the operators divided into several types.

1. **Assignment operator [=]: It copies the value on its right side into the variable on its left side. In assignment operation the left side operand**

should be a variable i.e. constant or expressions on left side not allowed.

Eg:

a=10;

b=1.2;

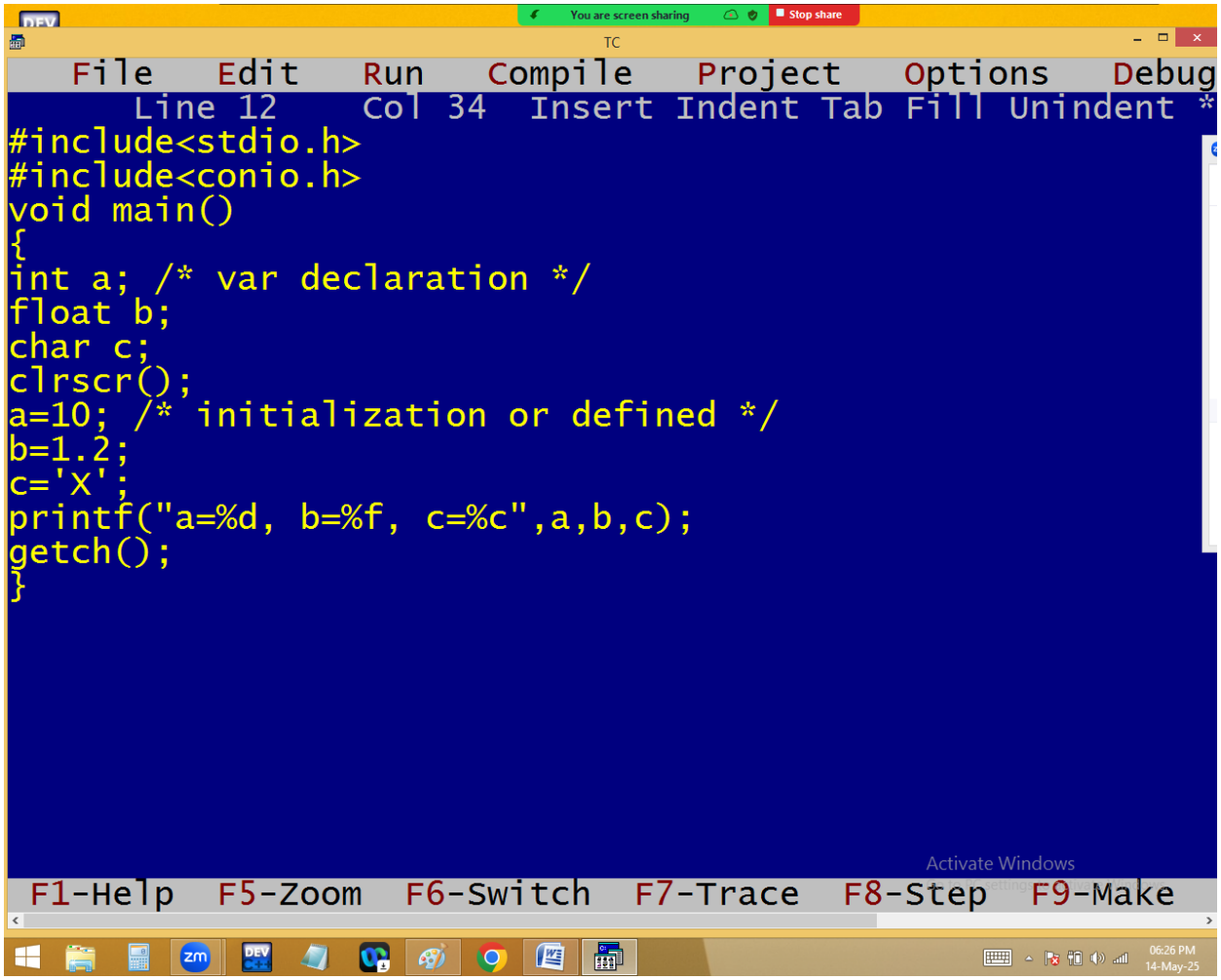
c='X';

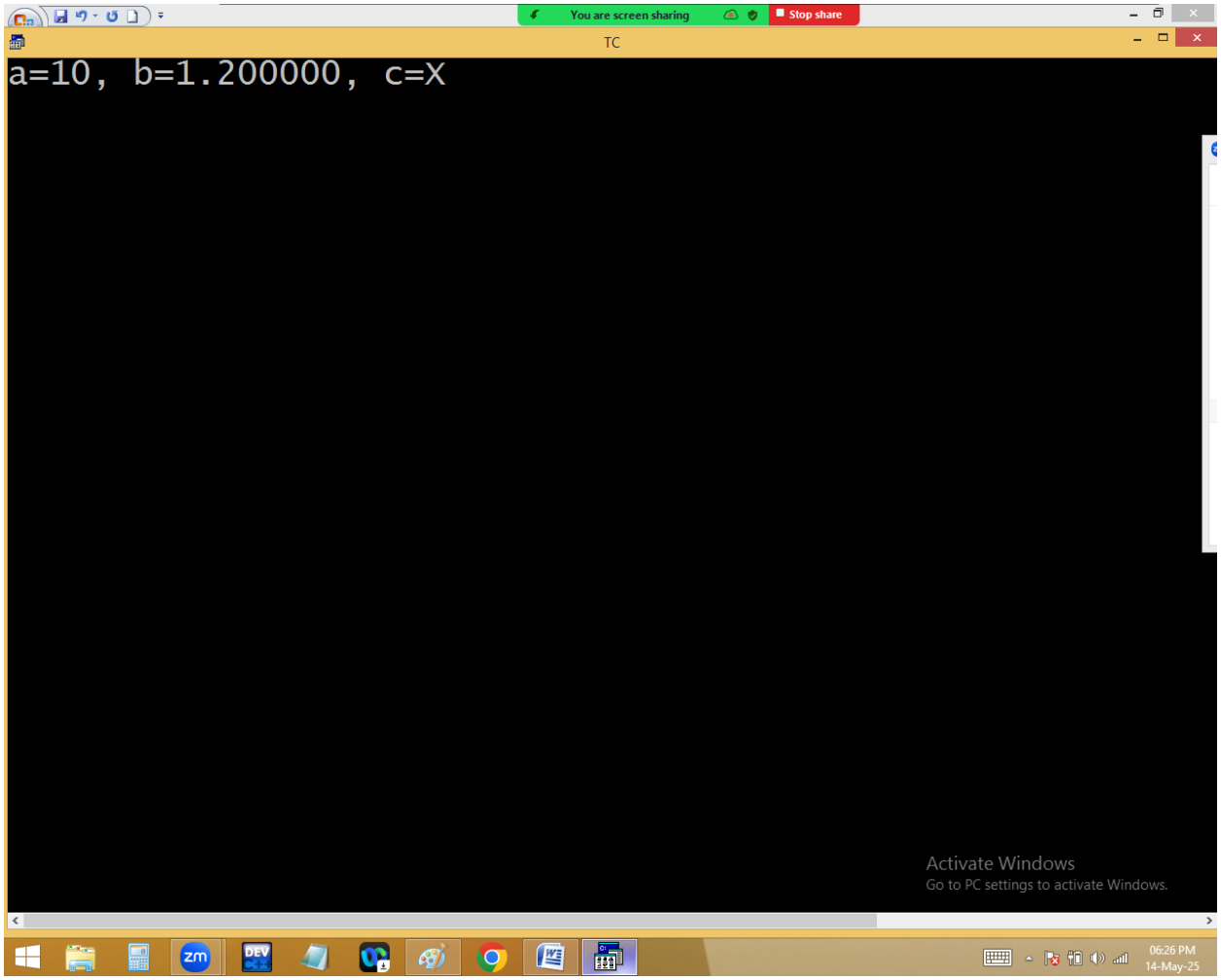
d="raj"; ➔ **Error because of raj is a string**

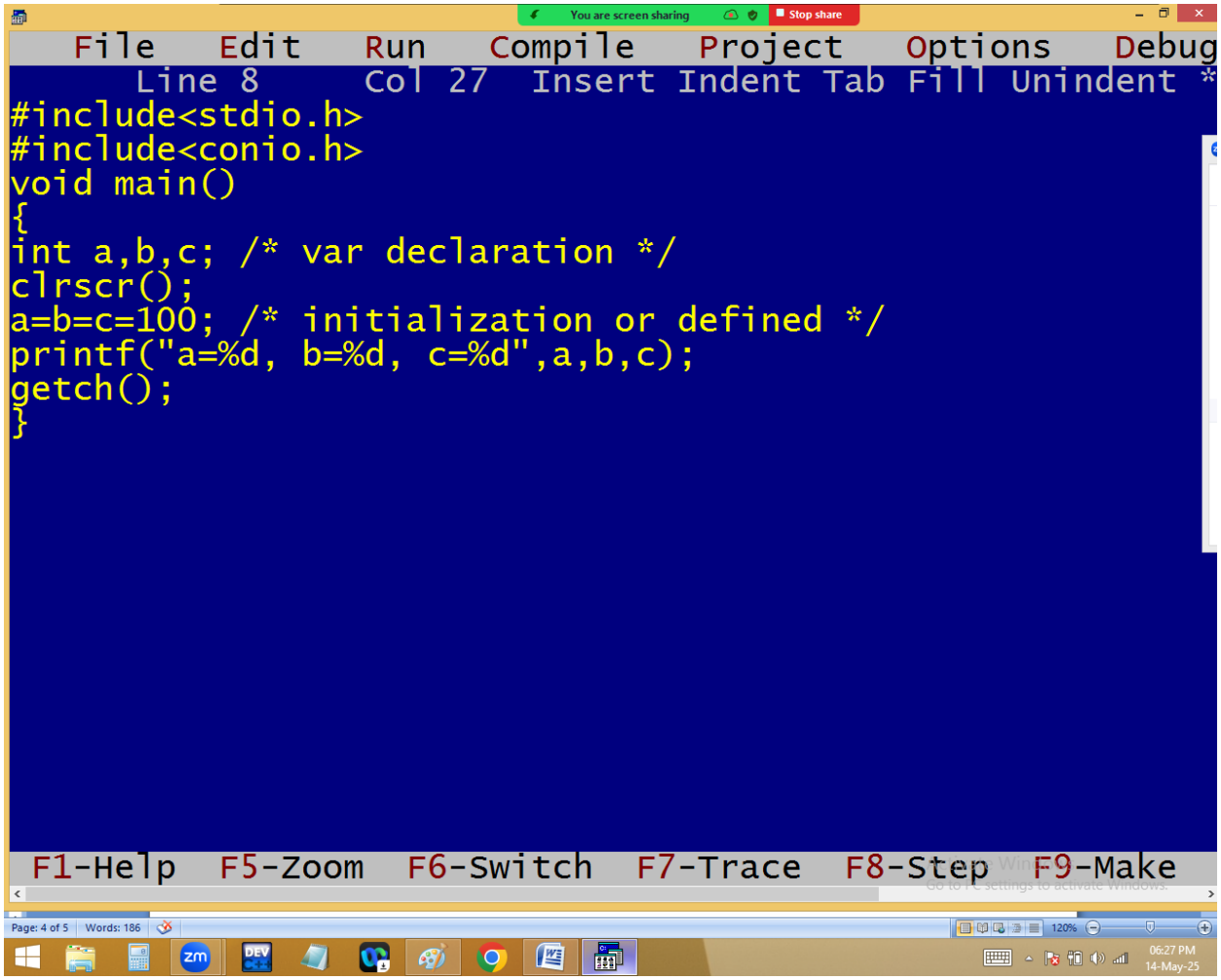
10=20; ➔ **Error because of 10 is a constant.**

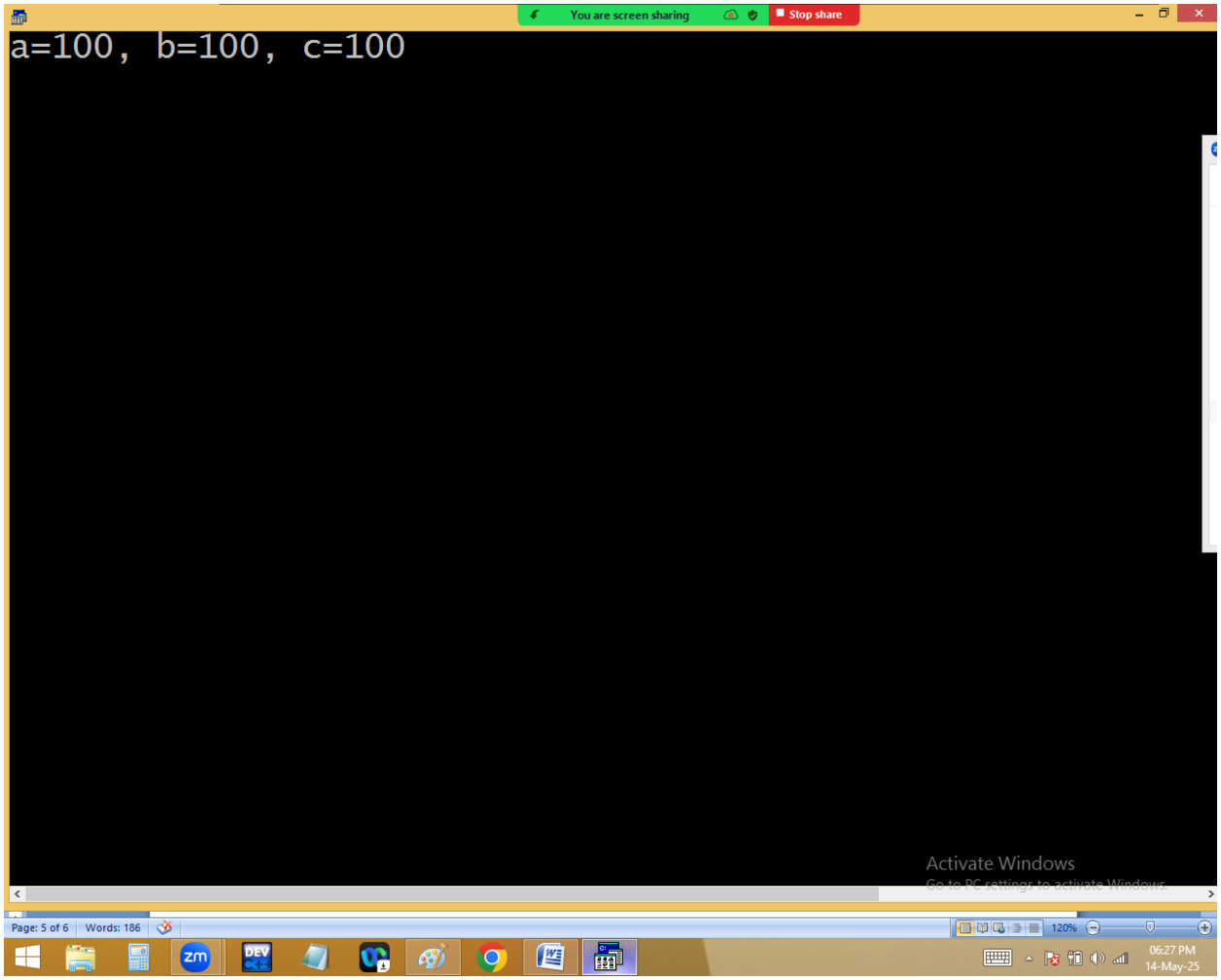
10+20=30 ➔ **30=30** ➔ **Error because of 10+20 gives 30 which is a constant**

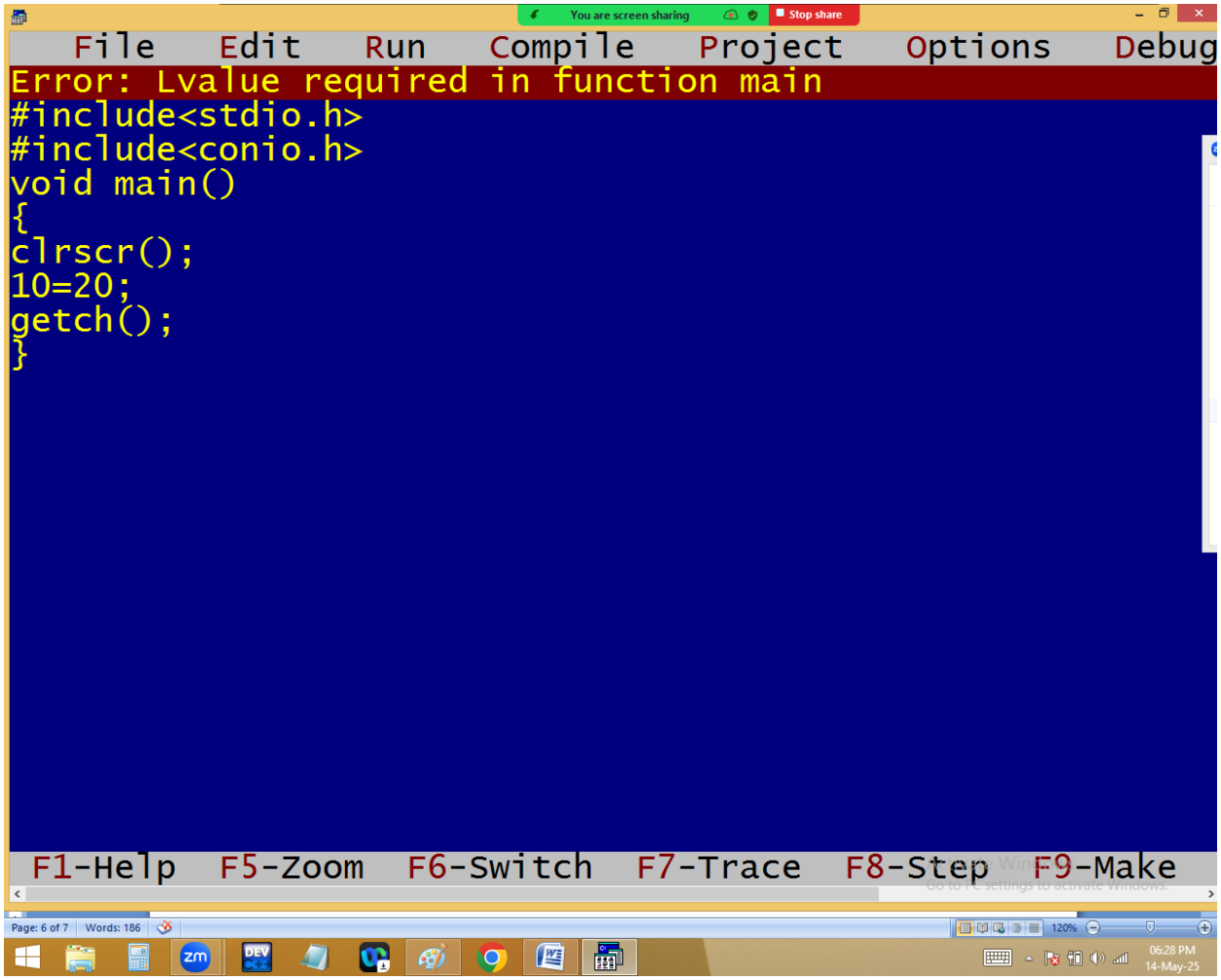
a=b=c=100;

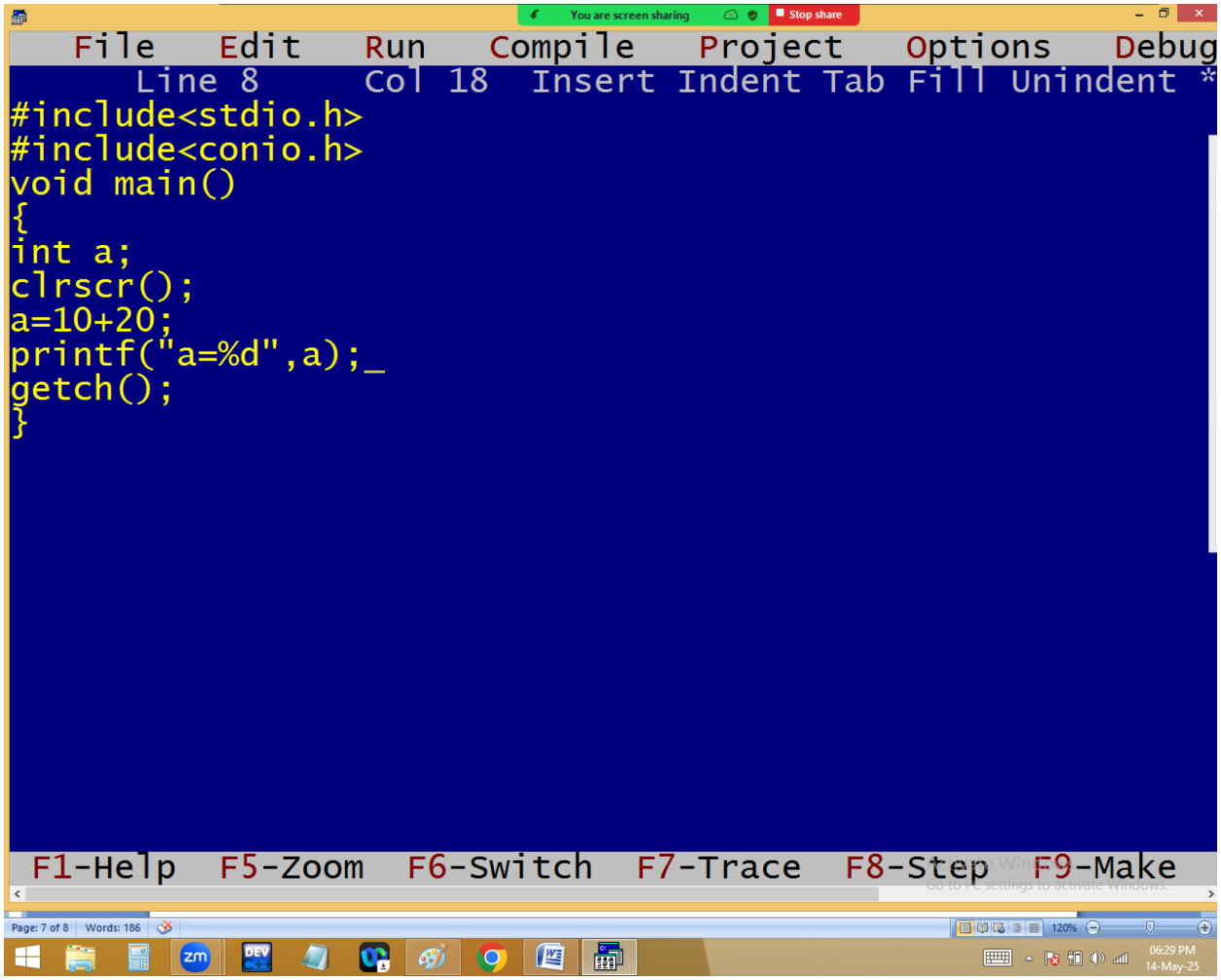

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TC
File Edit Run Compile Project Options Debug
Line 12 Col 34 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* var declaration */
float b;
char c;
clrscr();
a=10; /* initialization or defined */
b=1.2;
c='X';
printf("a=%d, b=%f, c=%c",a,b,c);
getch();
}
Activate Windows
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
06:26 PM
14-May-25

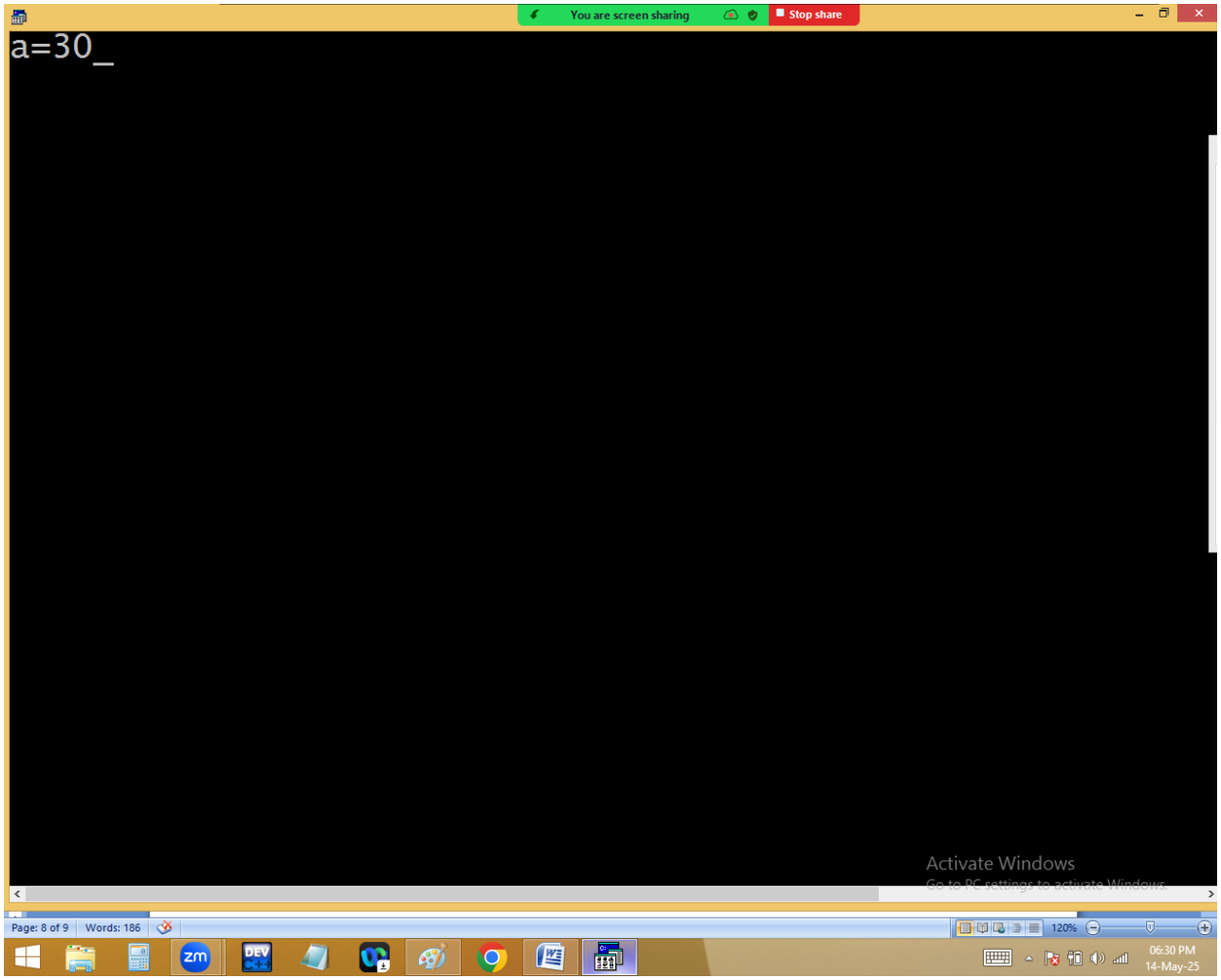

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TC
a=10, b=1.200000, c=X
Activate Windows
Go to PC settings to activate Windows.
06:26 PM
14-May-25

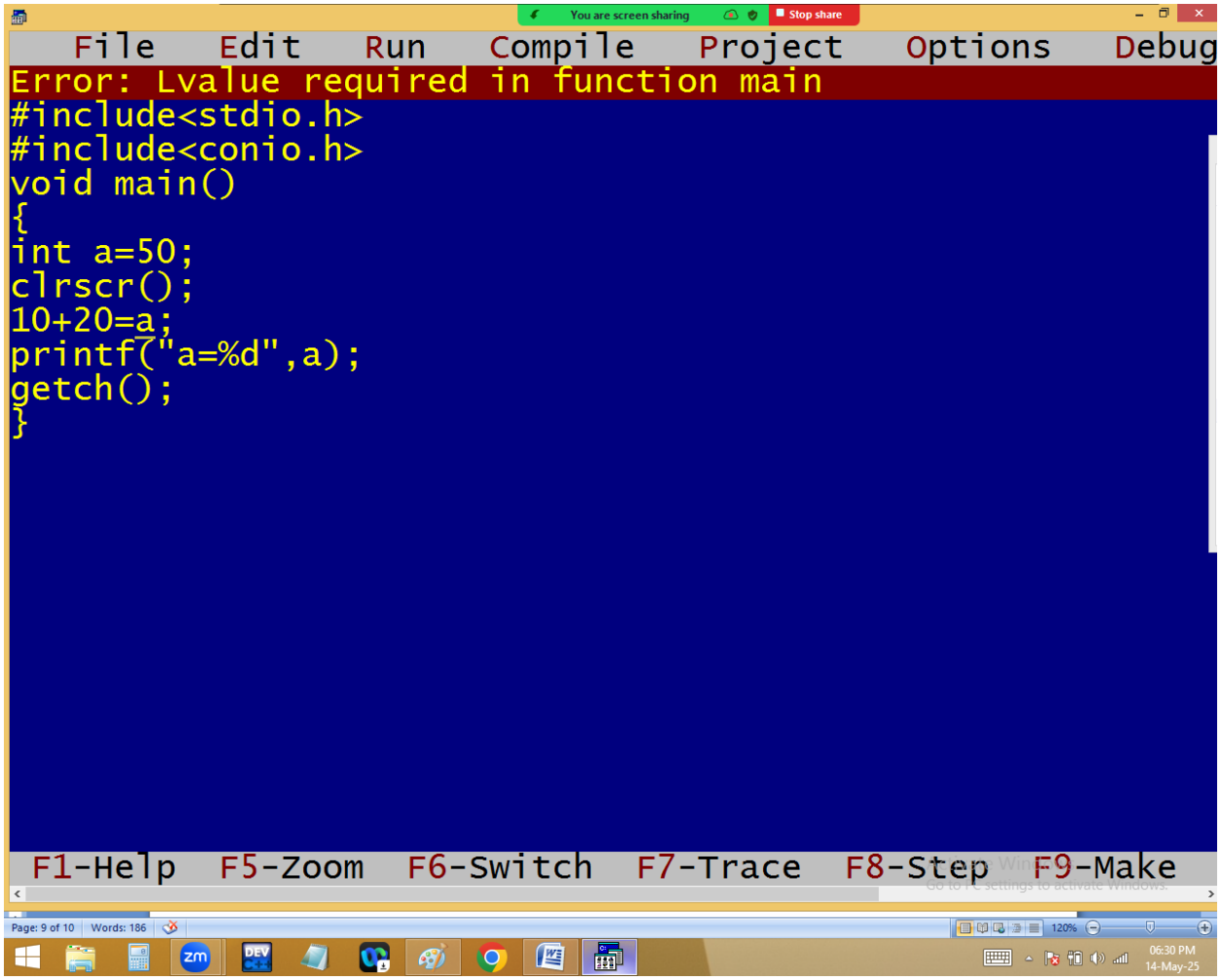

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File Edit Run Compile Project Options Debug
Line 8 Col 27 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c; /* var declaration */
clrscr();
a=b=c=100; /* initialization or defined */
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 4 of 5
Words: 186
120%
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14-May-25

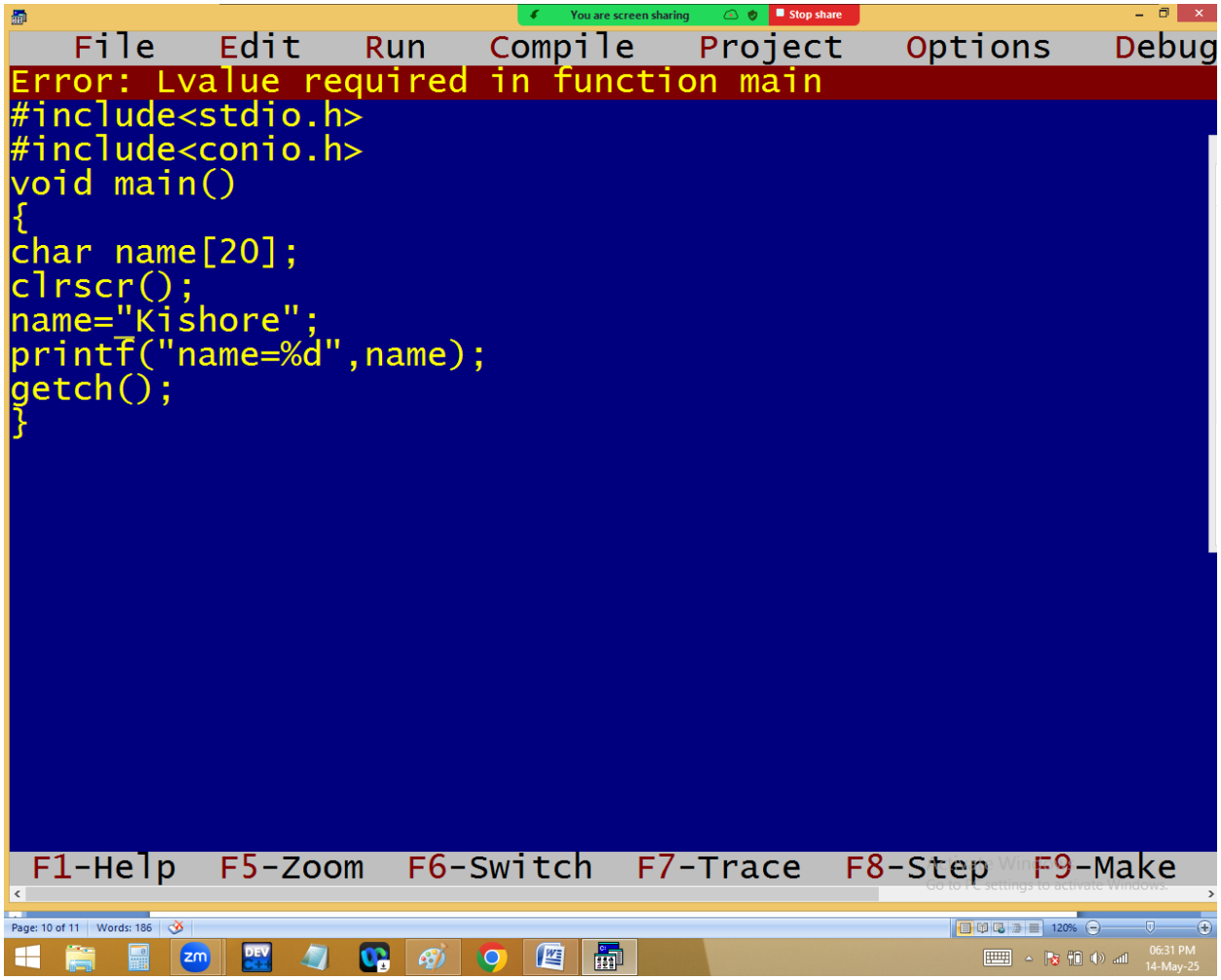

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a=100, b=100, c=100
Activate Windows
Page: 5 of 6
Words: 186
120%
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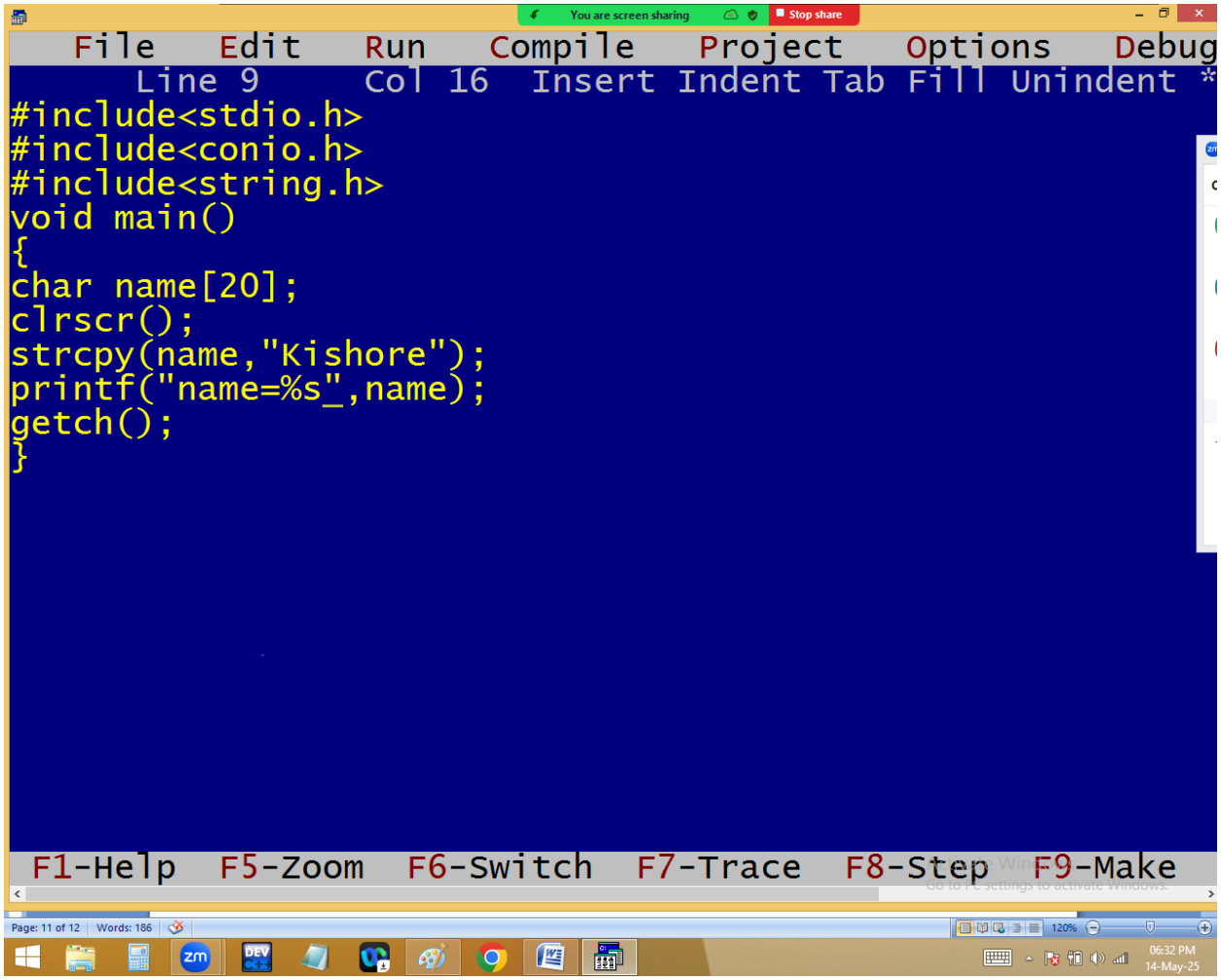

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File Edit Run Compile Project Options Debug
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
10=20;
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 6 of 7
Words: 186
120%
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14-May-25

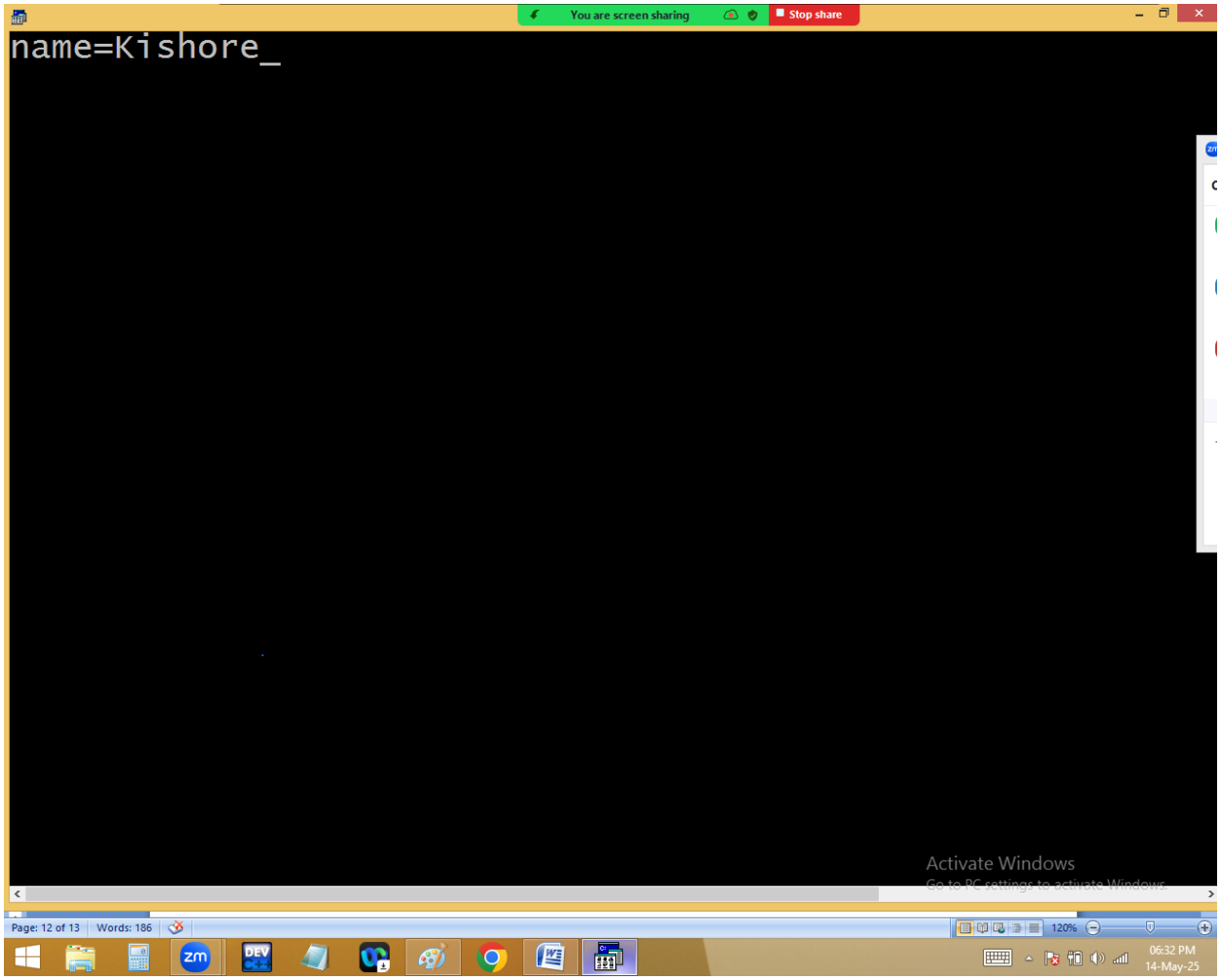

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File Edit Run Compile Project Options Debug
Line 8 Col 18 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
a=10+20;
printf("a=%d",a);_
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 7 of 8
Words: 186
120%
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Stop share
a=30_
Activate Windows
Page: 8 of 9
Words: 186
120%
06:30 PM
14-May-25


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File Edit Run Compile Project Options Debug
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a=50;
clrscr();
10+20=a;
printf("a=%d",a);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 9 of 10
Words: 186
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File Edit Run Compile Project Options Debug
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
char name[20];
clrscr();
name="Kishore";
printf("name=%d",name);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 10 of 11
Words: 186
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File Edit Run Compile Project Options Debug
Line 9 Col 16 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char name[20];
clrscr();
strcpy(name,"Kishore");
printf("name=%s",name);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 11 of 12
Words: 186
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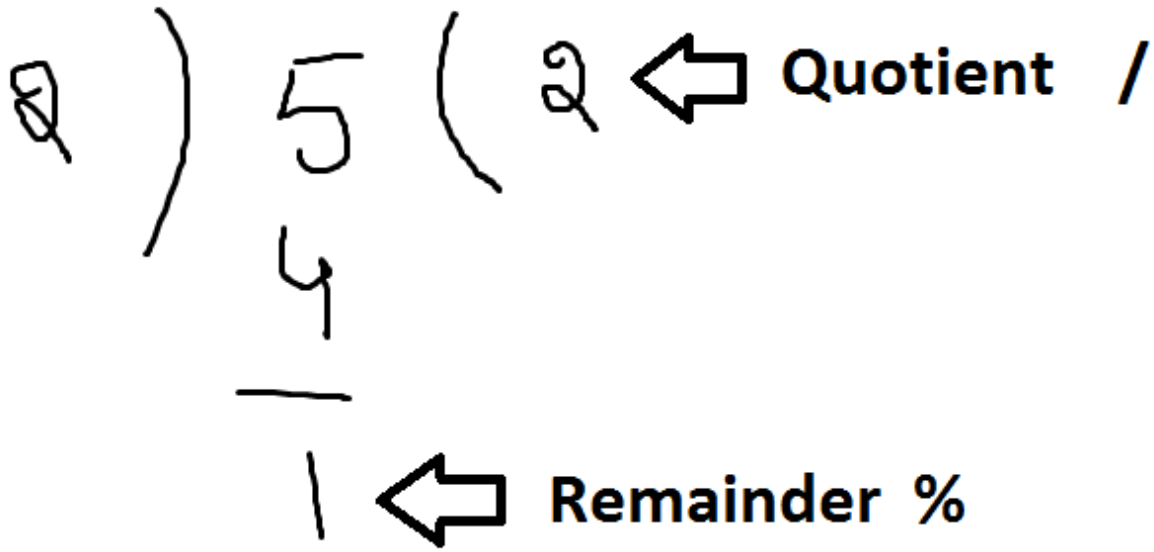
2. **Arithmetic operators [+, -, *, %, /]: They are used to perform mathematical operations.**

   Eg: a+b, a-b, a*b,…

   % - modules [Remainder]:

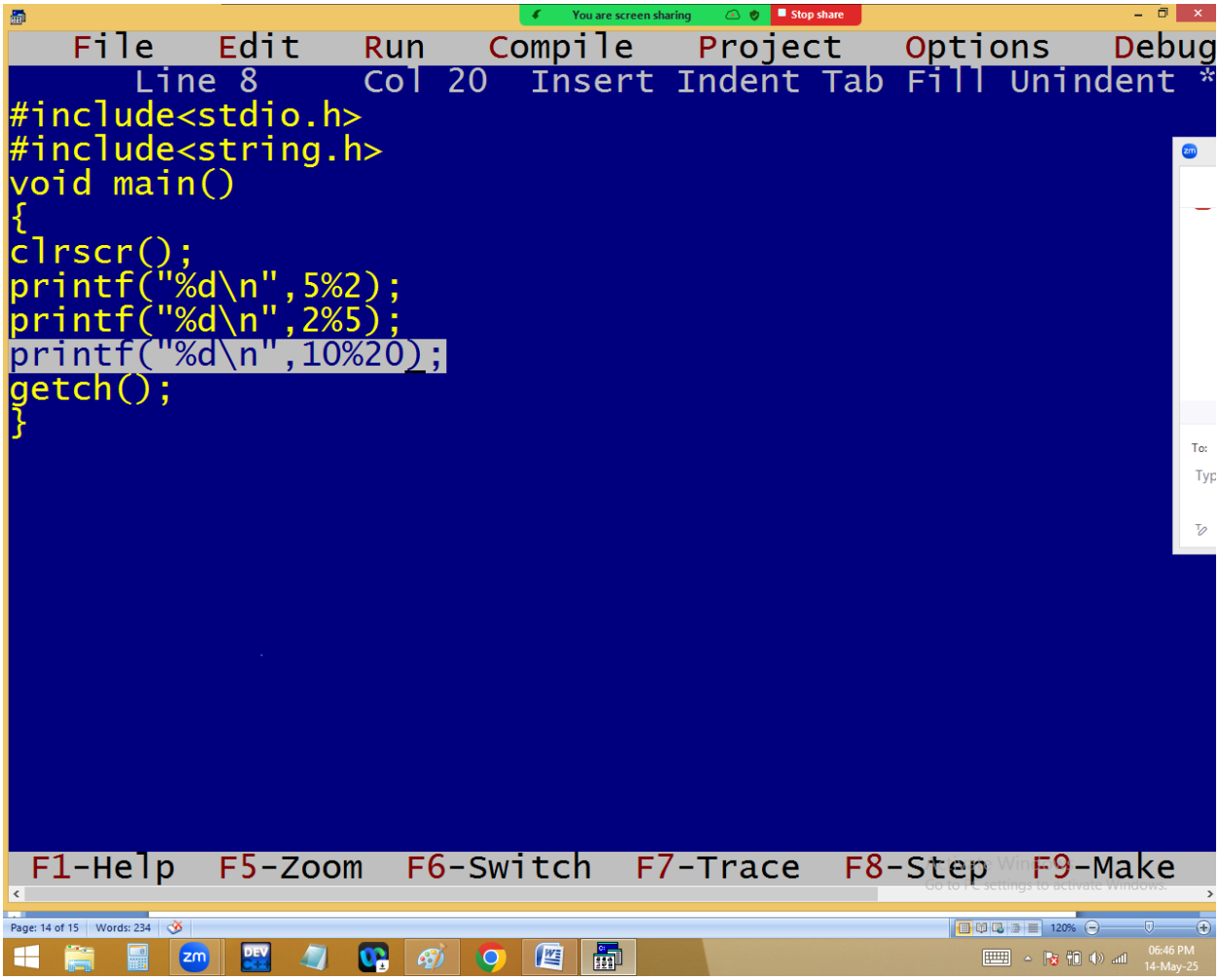
   Eg:

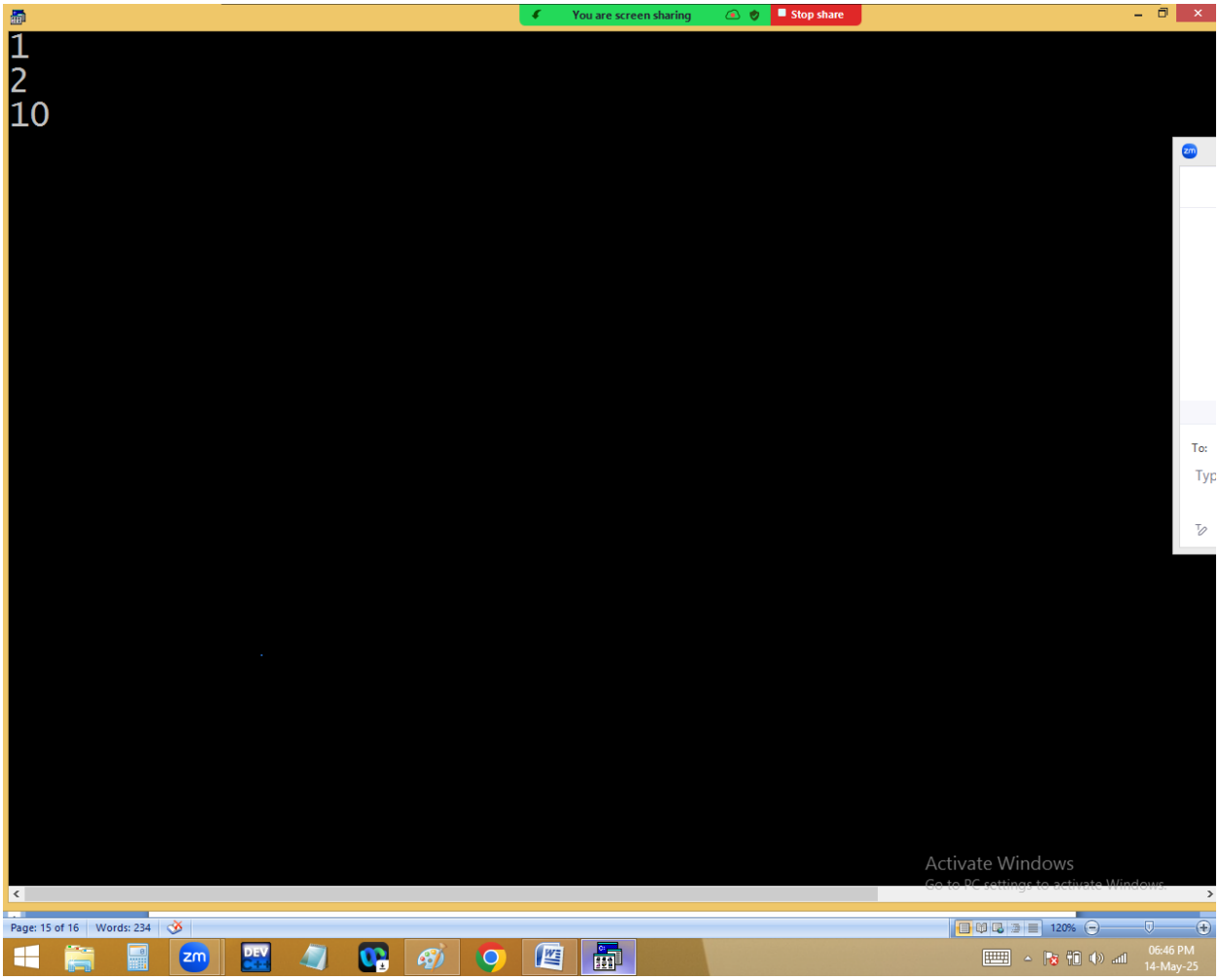
   5%2=1



2 % 5 = 2

Note: If the divisor is bigger than dividend then the dividend is the answer.

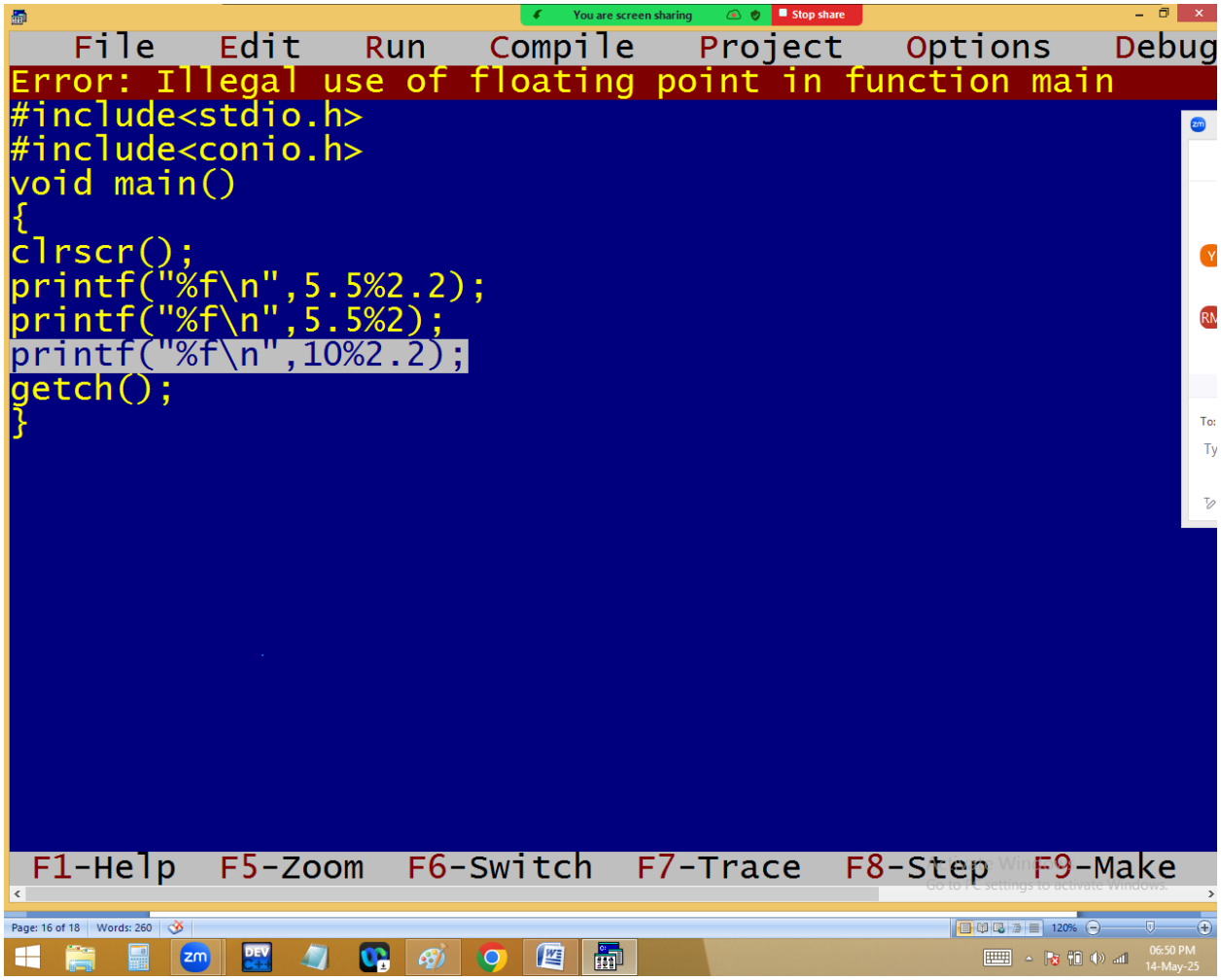

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File Edit Run Compile Project Options Debug
Line 8 Col 20 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<string.h>
void main()
{
clrscr();
printf("%d\n",5%2);
printf("%d\n",2%5);
printf("%d\n",10%20);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 14 of 15
Words: 234
120%
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14-May-25

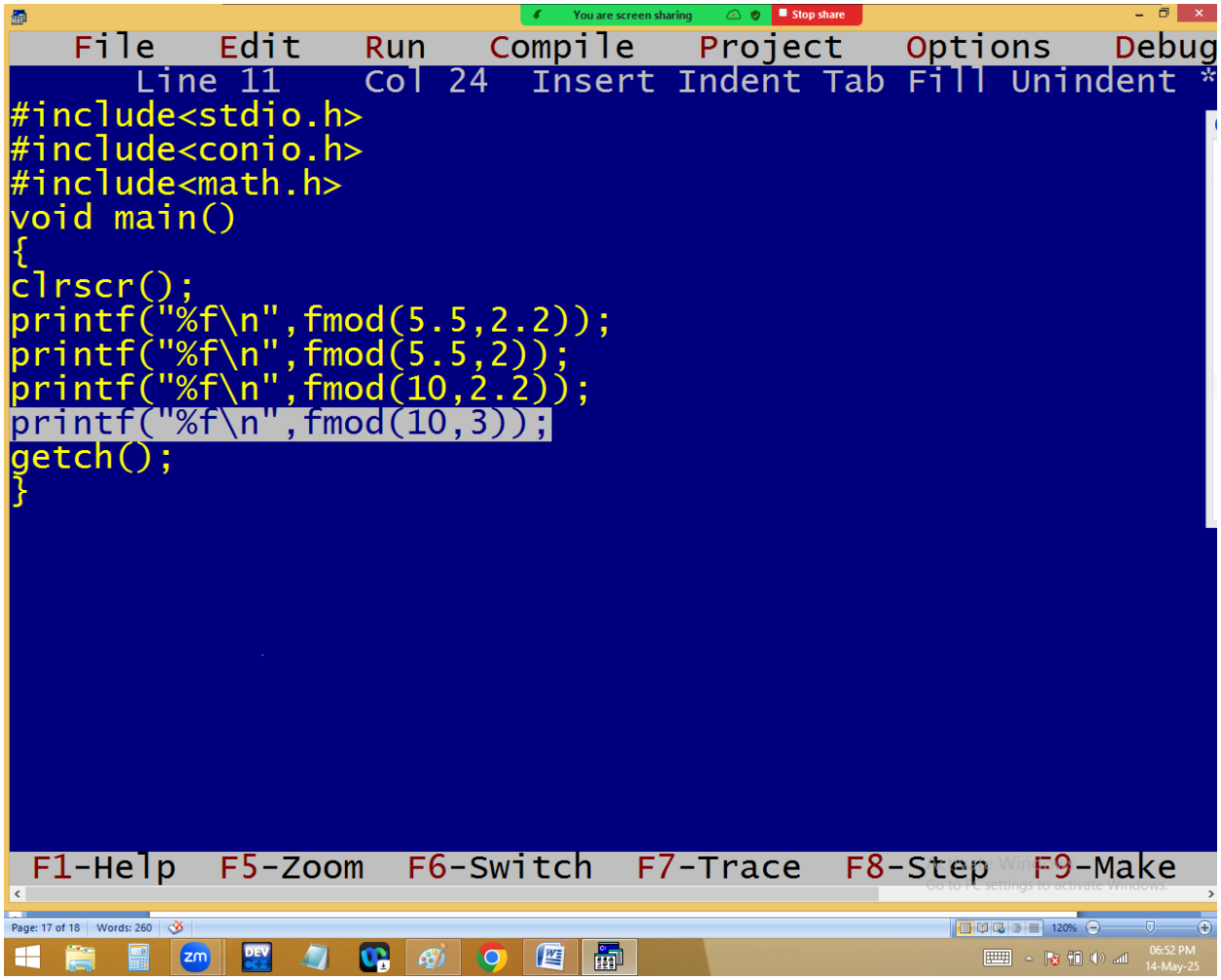


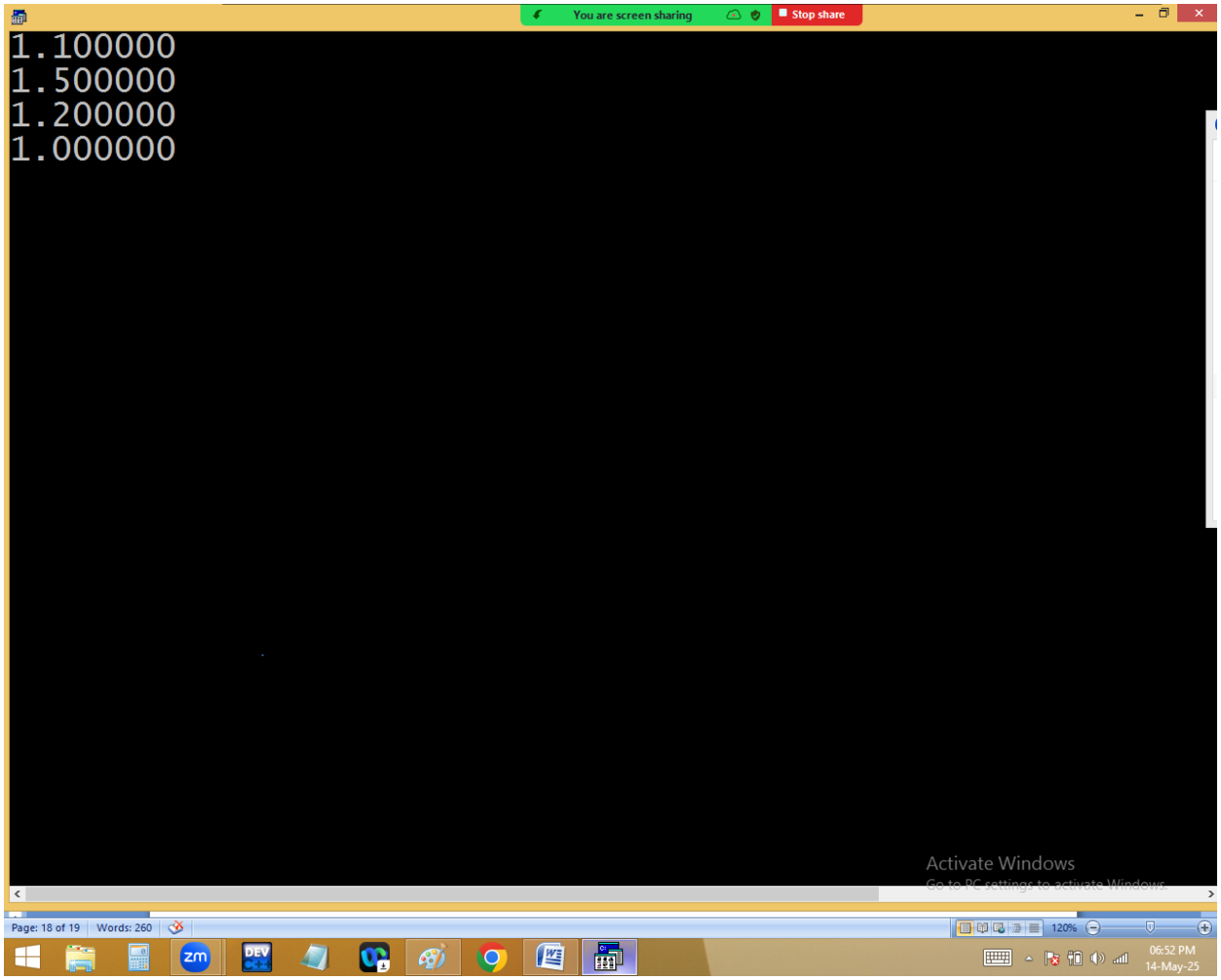
5.0%2.0=Error

Note: In C & C++ we can't do floating modules with % operator.

To do floating modules we have to use fmod() available in <math.h>


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File Edit Run Compile Project Options Debug
Error: Illegal use of floating point in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",5.5%2.2);
printf("%f\n",5.5%2);
printf("%f\n",10%2.2);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 16 of 18
Words: 260
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File Edit Run Compile Project Options Debug
Line 11 Col 24 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
clrscr();
printf("%f\n",fmod(5.5,2.2));
printf("%f\n",fmod(5.5,2));
printf("%f\n",fmod(10,2.2));
printf("%f\n",fmod(10,3));
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 17 of 18
Words: 260
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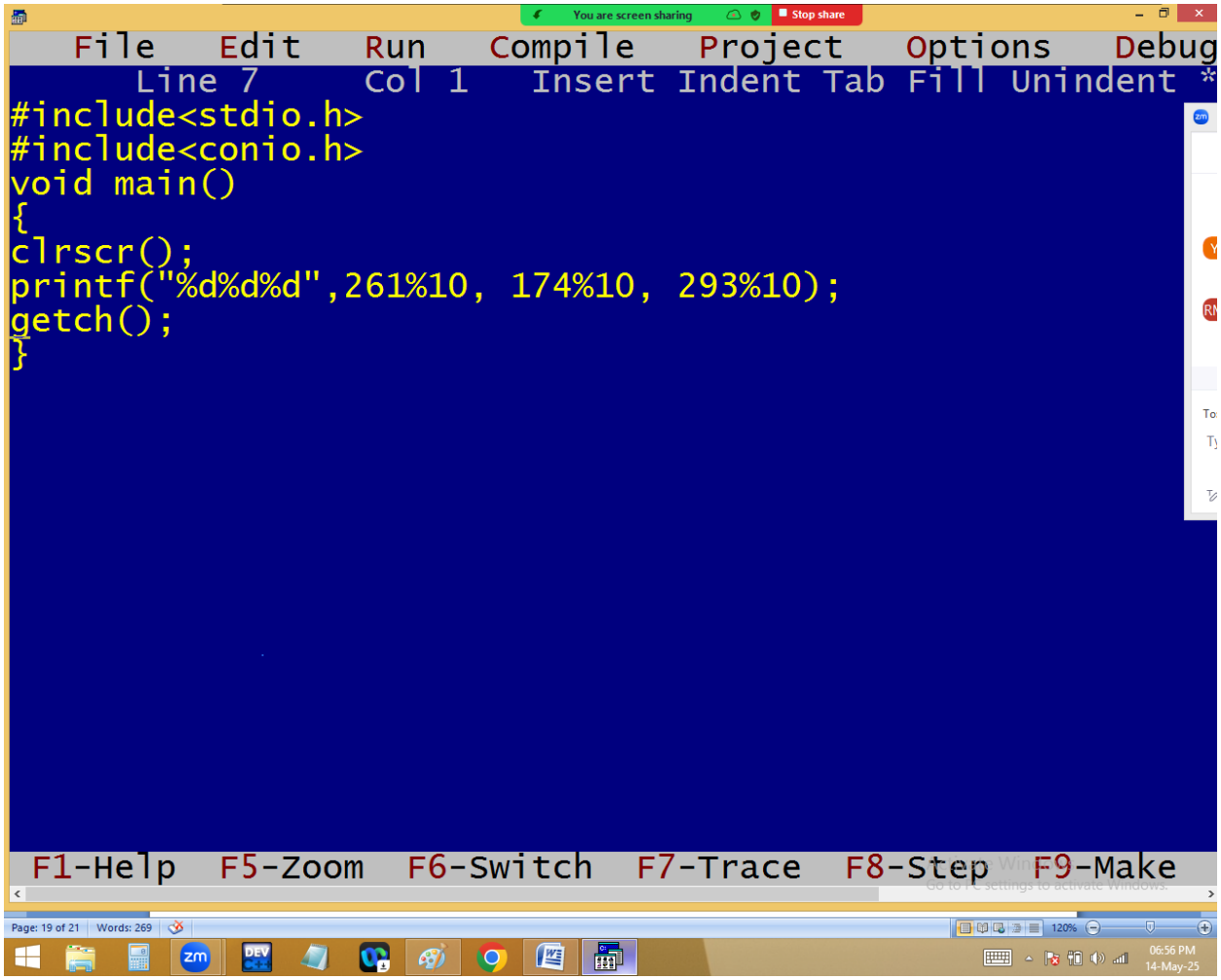


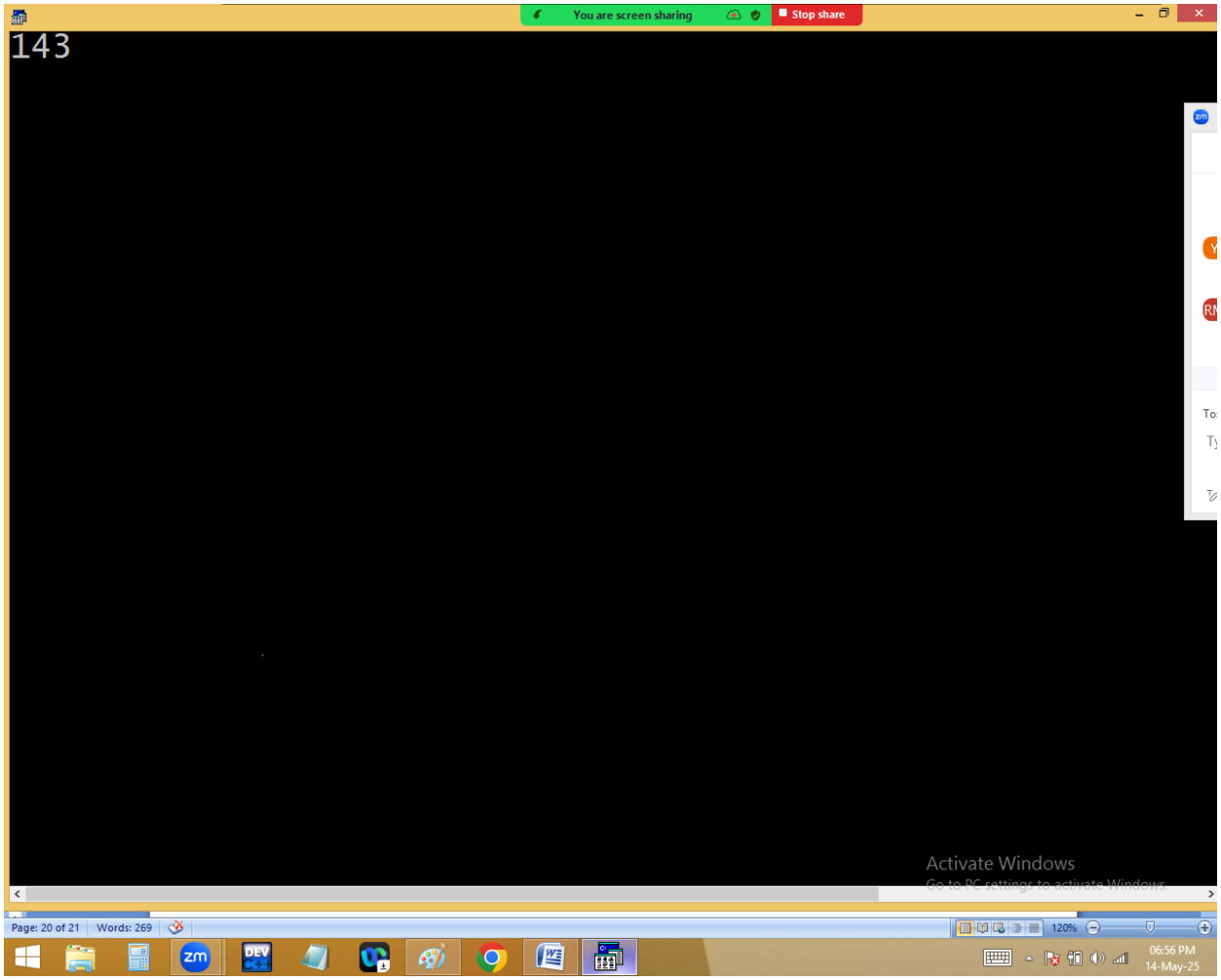
241%10=1

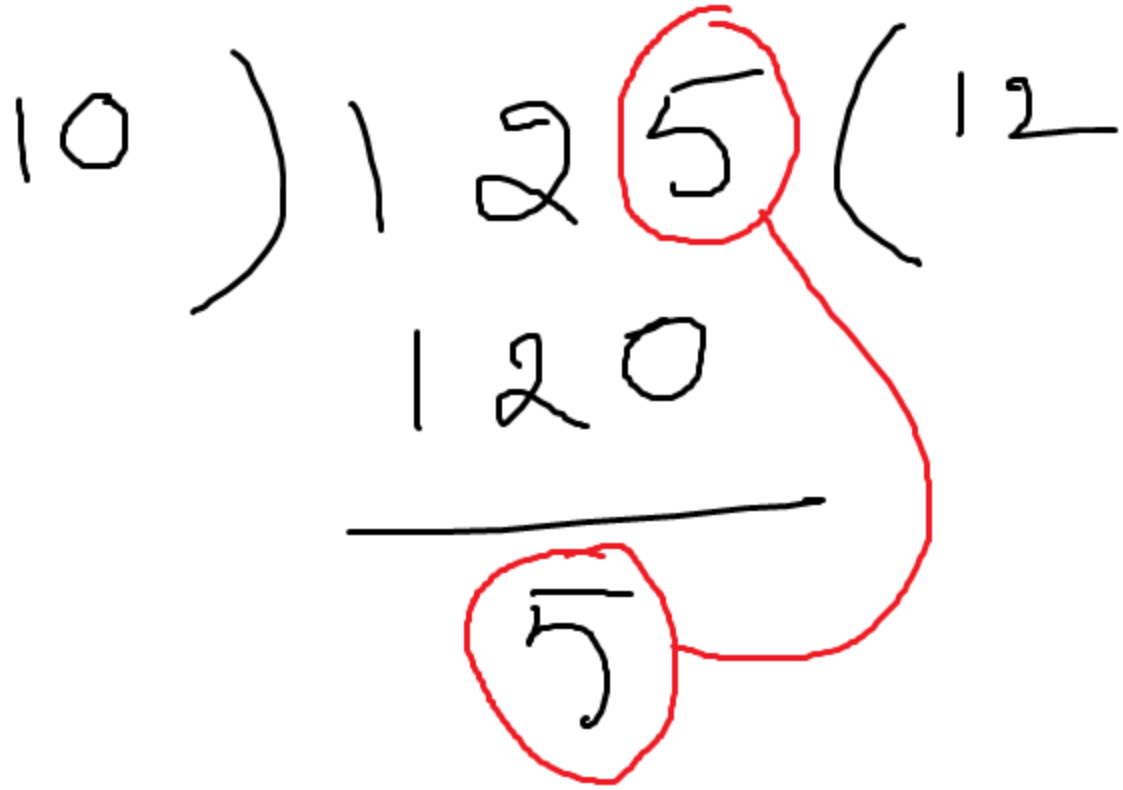
124%10=4

953%10=3

Note: Any no%10 gives last digit.


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File Edit Run Compile Project Options Debug
Line 7 Col 1 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d%d%d",261%10, 174%10, 293%10);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 19 of 21
Words: 269
120%
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143
Activate Windows
Page: 20 of 21
Words: 269
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/ - division [quotient]:

5/2=2

Note: In c & C++ both operands are integers then result also integer. If anyone or both are floats result also float.

5.0/2=2.500000

5/2.0=2.500000

5.0/2.0=2.500000

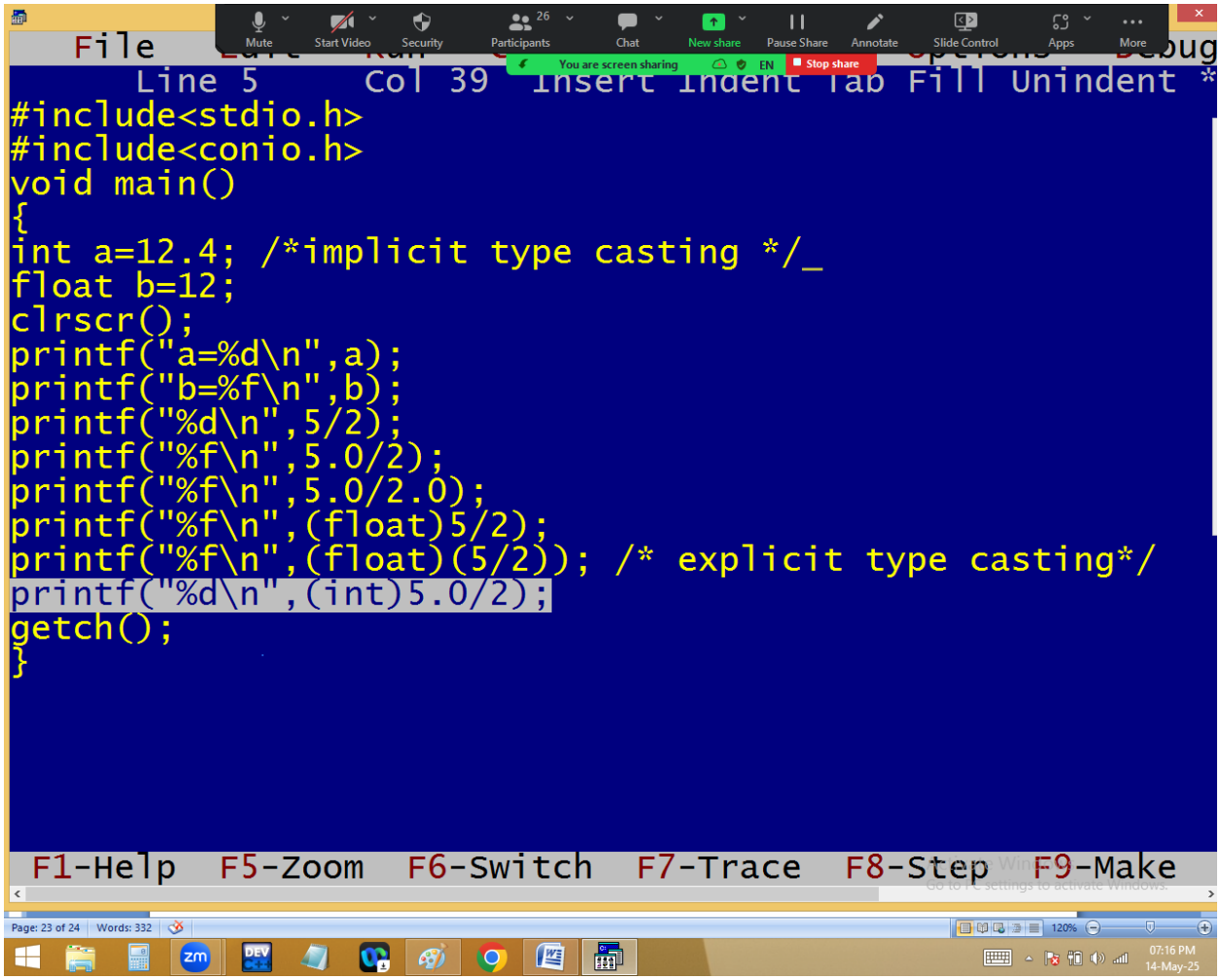
(float)5/2=2.500000 /* explicit type casting */

(int)5.0/2=2 /* explicit type casting */

Float a = 10; ➔ 10.000000 ➔ implicit type casting

Int b=10.4; ➔ 10 ➔ implicit type casting

(float)(5/2)=2.000000


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File
Line 5 Col 39 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int a=12.4; /*implicit type casting */_
float b=12;
clrscr();
printf("a=%d\n",a);
printf("b=%f\n",b);
printf("%d\n",5/2);
printf("%f\n",5.0/2);
printf("%f\n",5.0/2.0);
printf("%f\n",(float)5/2);
printf("%f\n",(float)(5/2)); /* explicit type casting*/
printf("%d\n",(int)5.0/2);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 23 of 24
Words: 332
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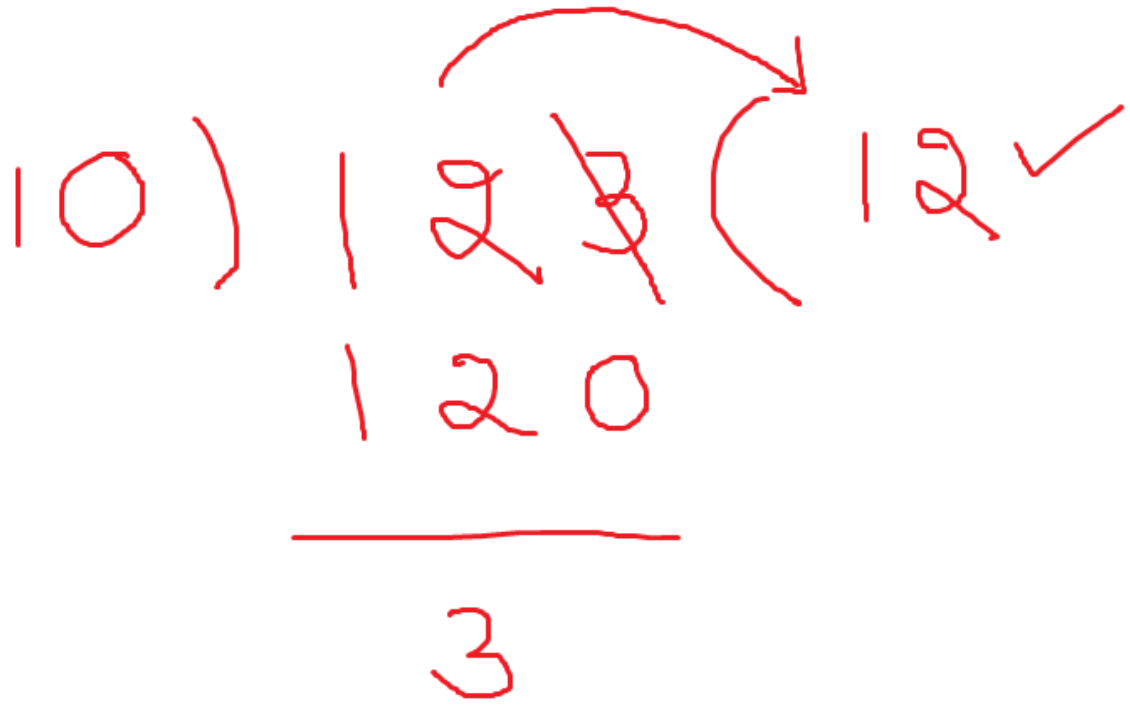


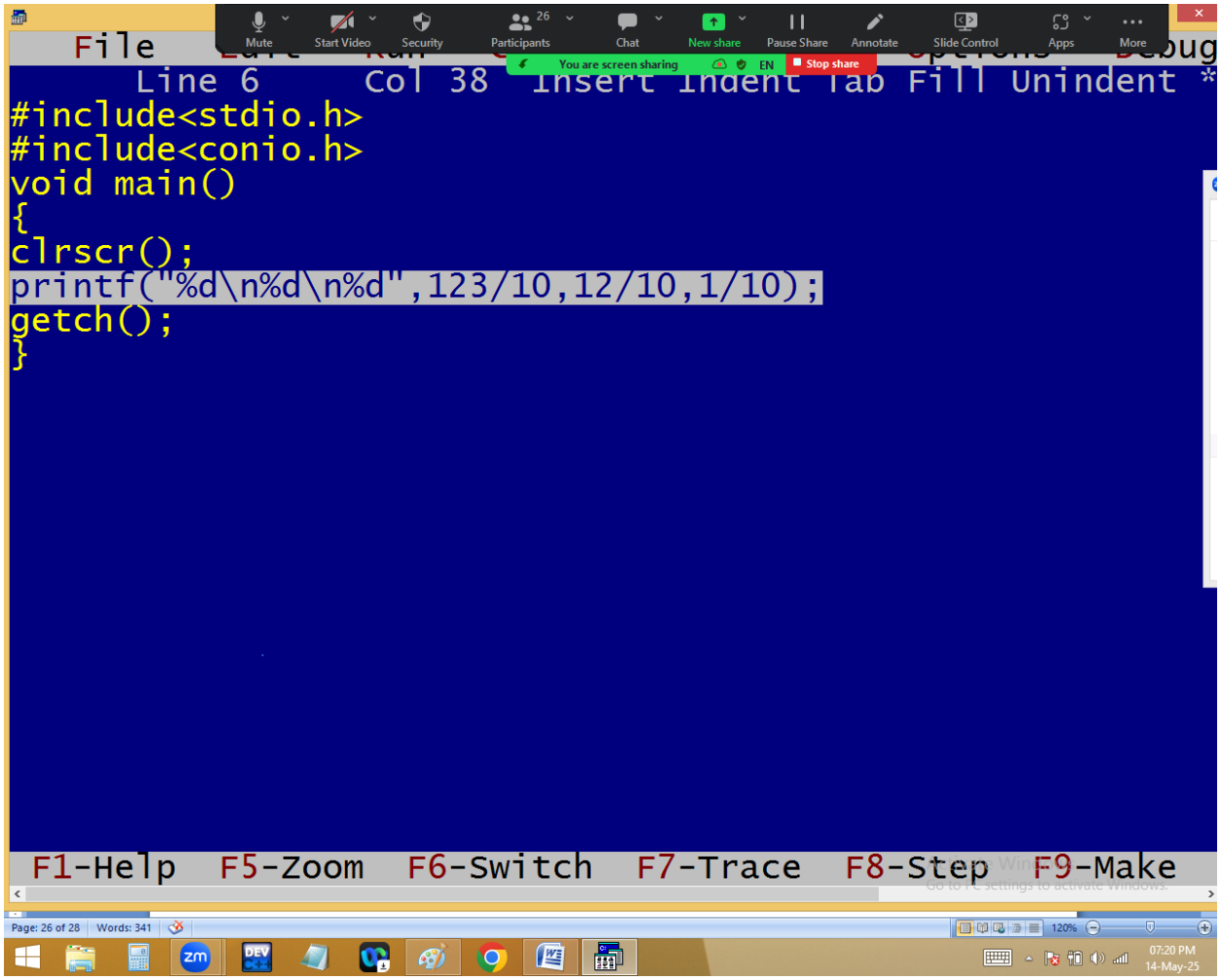
123/10=12

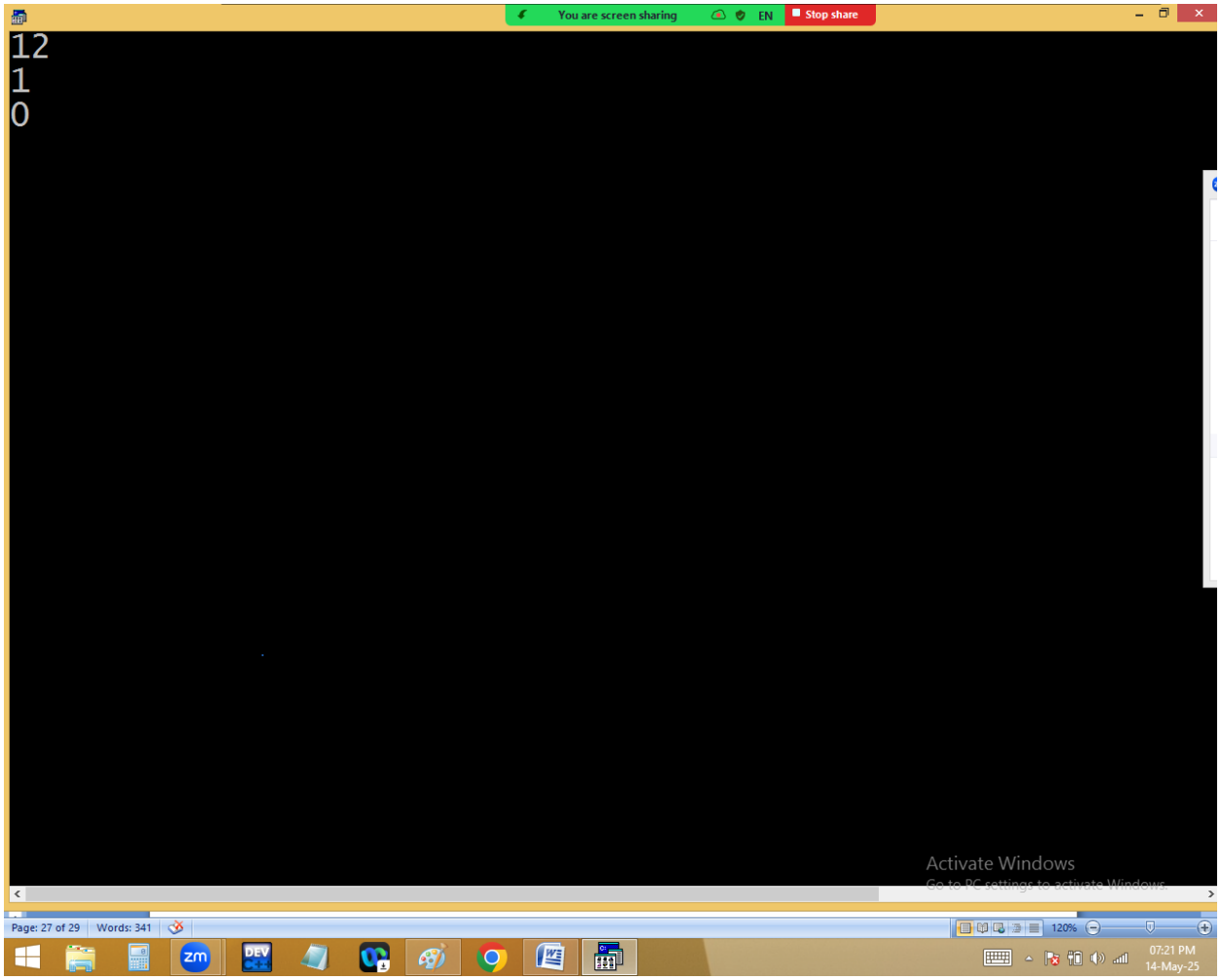
12/10=1

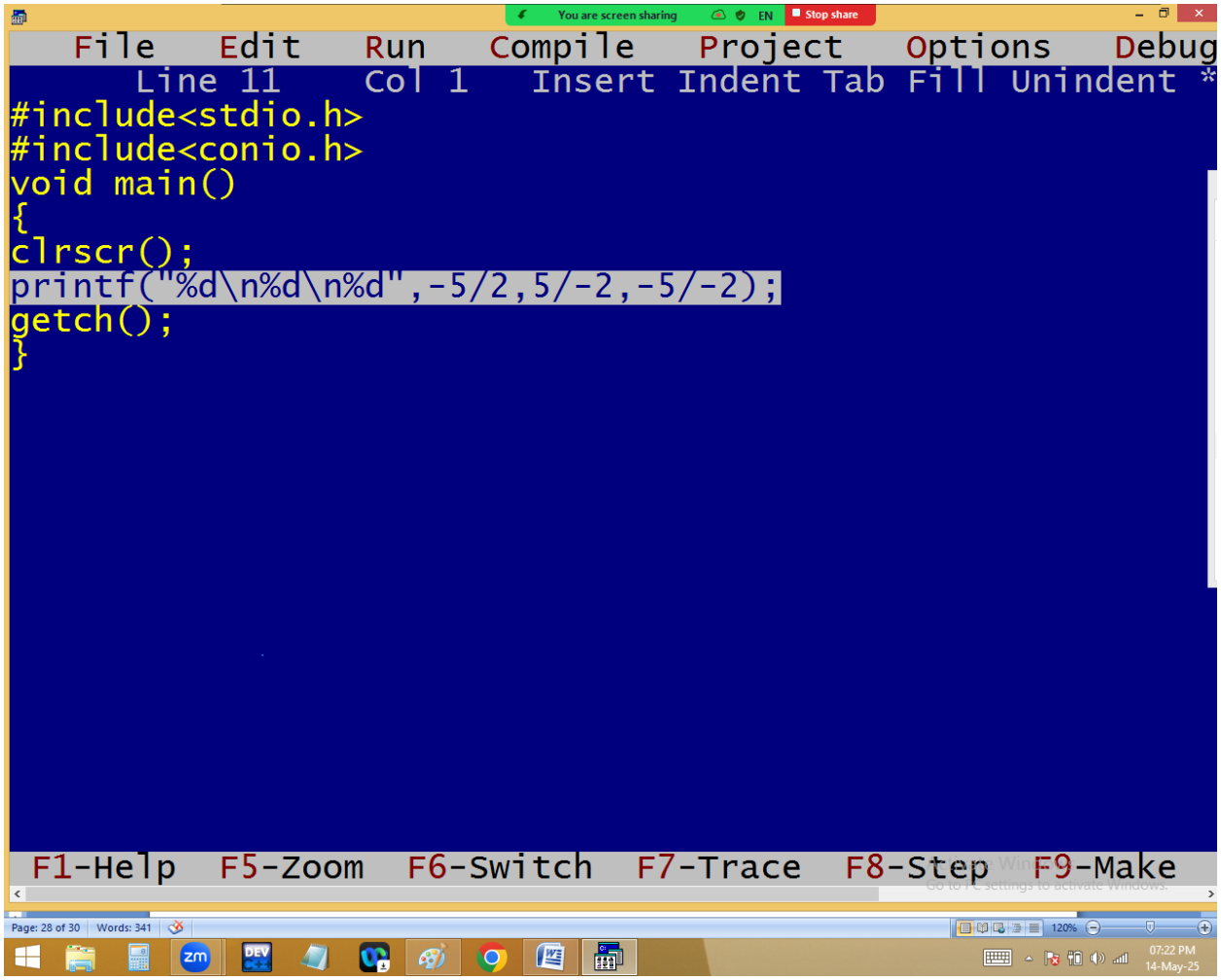
1/10=0

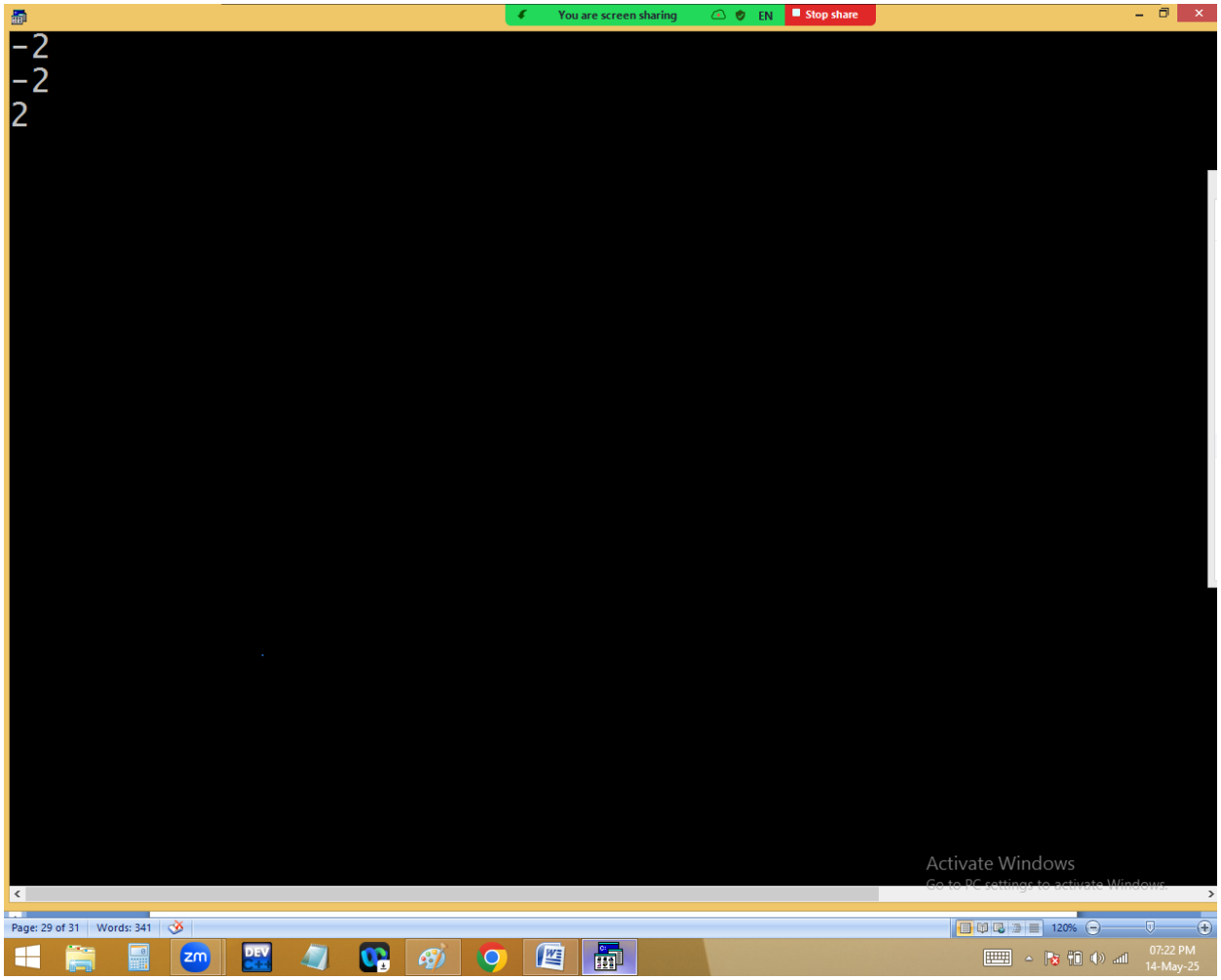
Note: Any no/10 removes last digit.


10) 123 (12
120
3


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EN
Stop share
File
Line 6 Col 38 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n%d\n%d",123/10,12/10,1/10);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 26 of 28
Words: 341
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12
1
0
Activate Windows
Page: 27 of 29
Words: 341
120%
07:21 PM
14-May-25


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File Edit Run Compile Project Options Debug
Line 11 Col 1 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n%d\n%d",-5/2,5/-2,-5/-2);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 28 of 30
Words: 341
120%
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14-May-25



Operator precedence / Operator priority

(ASSOCIATION OF OPERATORS)

1. ( )
2. +, - , ! (sign operators, unary operators)
3. ++, -- (pre increment & decrement)
4. *, / , %
5. +, - (Binary)

6. = =, !=
7. &&
8. ||
9. **? :** (ternary operator)
10. =
11. ++, -- (Post increment & decrement)
12. , (comma)